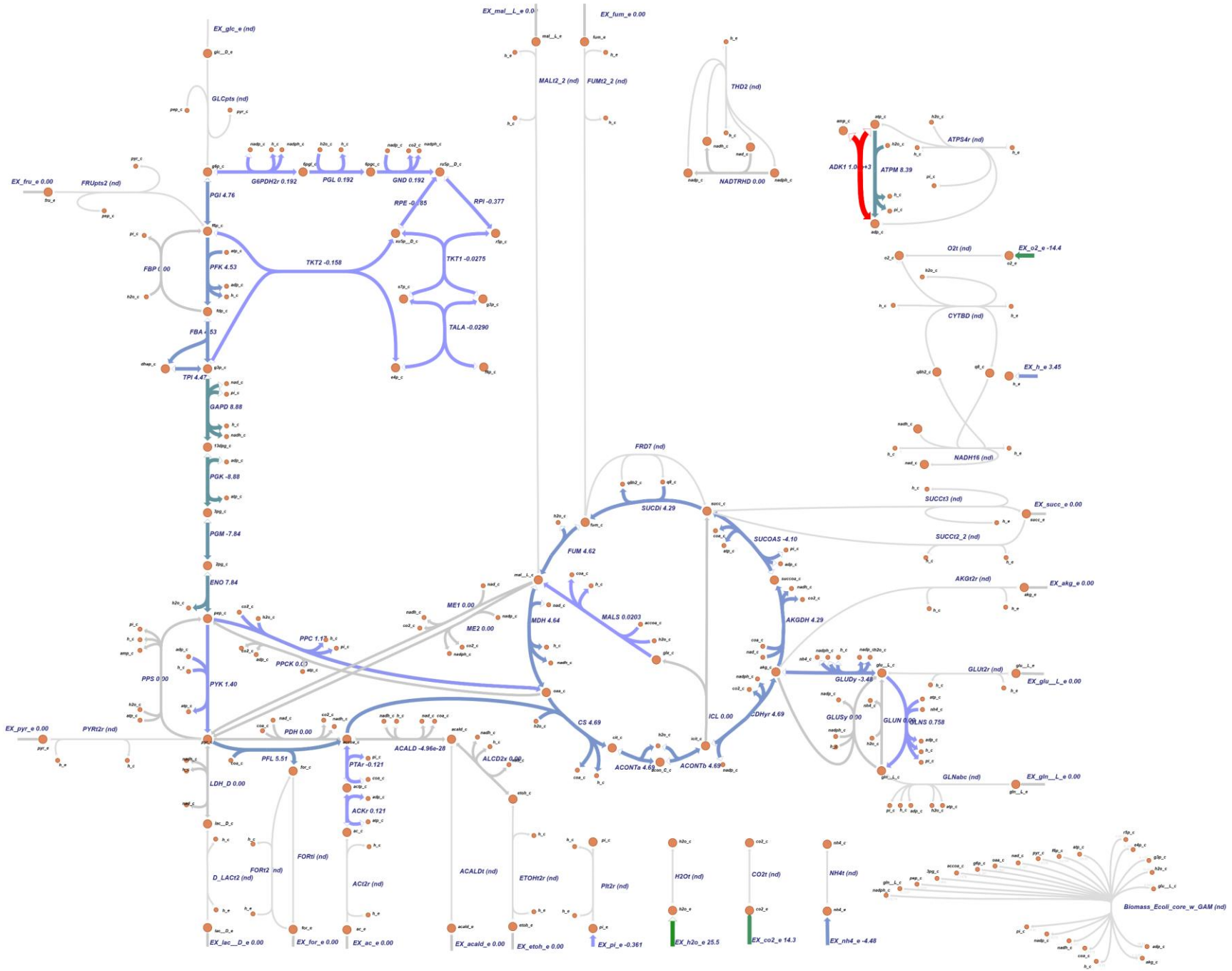
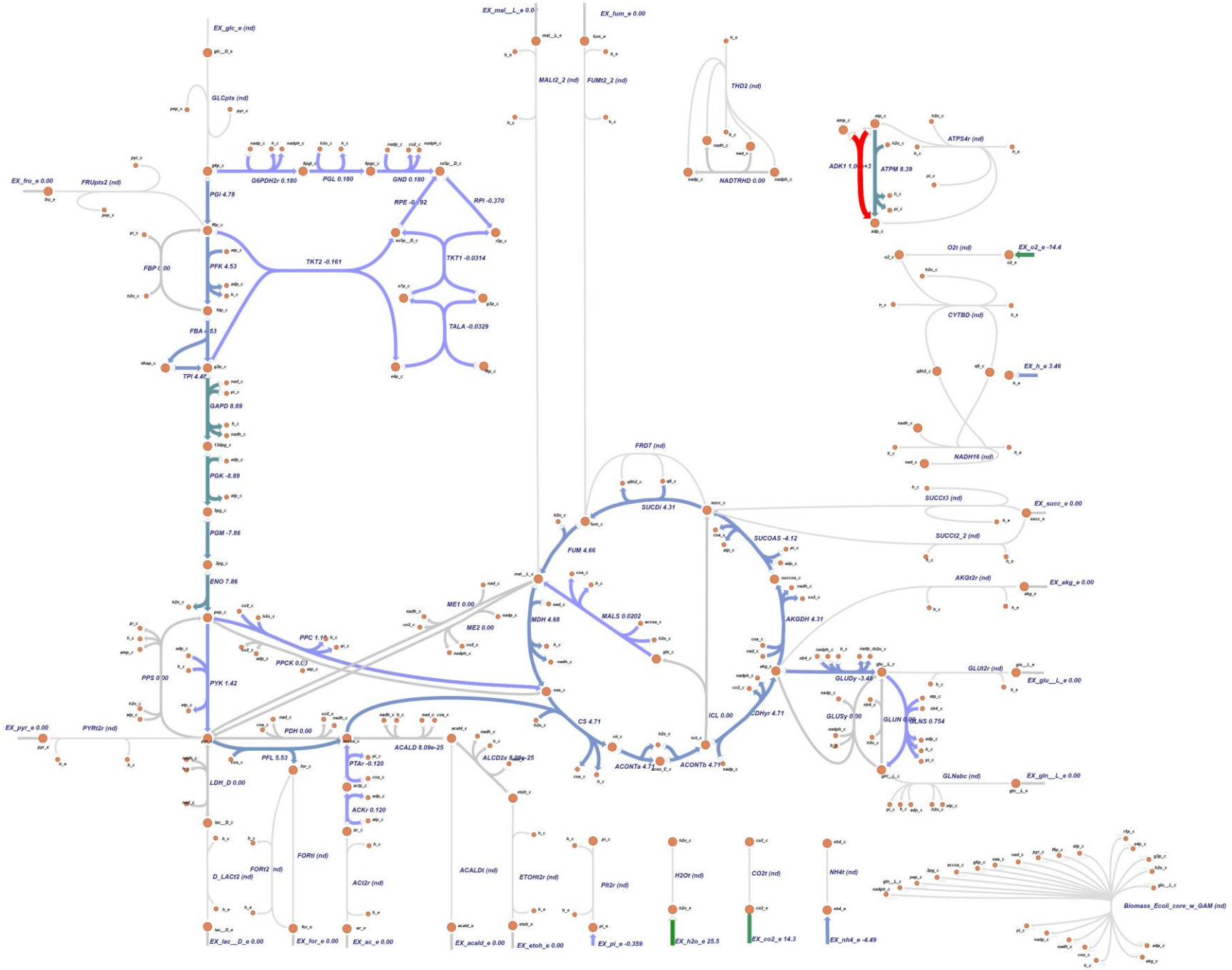
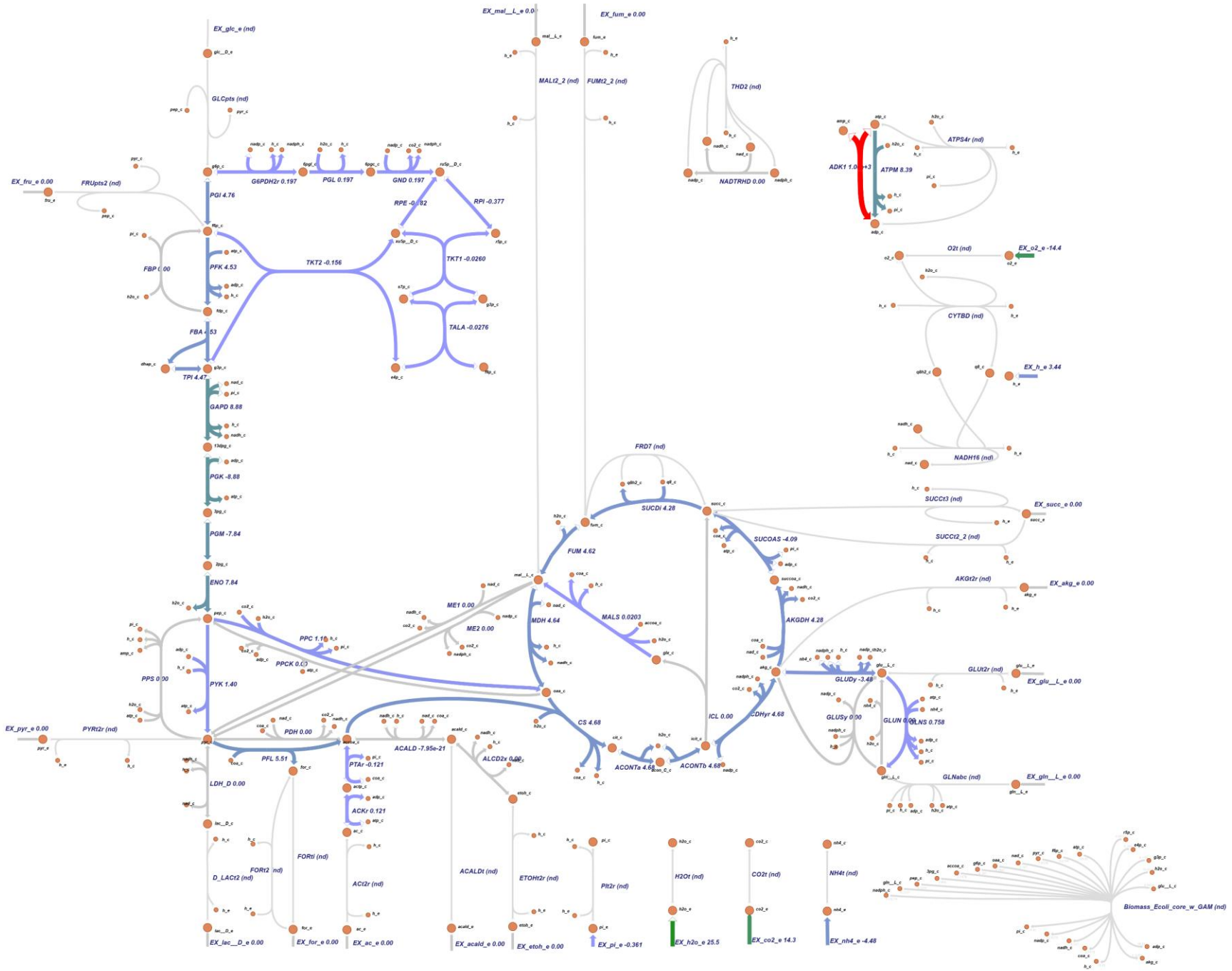
ΔlppA early


EX_glc_e (nd)
GLCpts (nd)
EX_fru_e 0.00
FRUpts2 (nd)
G6PDH2r 0.192
PGL 0.192
GND 0.192
PGI 4.76
RPE -0.085
RPI -0.377
TKT2 -0.158
TKT1 -0.0275
TALA -0.0290
PFK 4.53
FBP 0.00
FBA 4.53
TPI 4.47
GAPD 8.88
PGK -8.88
PGM -7.84
ENO 7.84
PPC 1.13
PPCK 0.00
PPS 0.00
PYK 1.40
EX_pyr_e 0.00
PYRt2r (nd)
PDH 0.00
PFL 5.51
LDH_D 0.00
D_LACt2 (nd)
FORt2 (nd)
FORti (nd)
ACt2r (nd)
PTAr -0.121
ACKr 0.121
ACALD -4.96e-28
ALCD2x 0.00
ACALDt (nd)
ETOHt2r (nd)
Pit2r (nd)
H2Ot (nd)
CO2t (nd)
NH4t (nd)
EX_lac__D_e 0.00
EX_for_e 0.00
EX_ac_e 0.00
EX_acald_e 0.00
EX_etoh_e 0.00
EX_pi_e -0.361
EX_h2o_e 25.5
EX_co2_e 14.3
EX_nh4_e -4.48
EX_mal__L_e 0.00
EX_fum_e 0.00
MALt2_2 (nd)
FUMt2_2 (nd)
THD2 (nd)
NADTRHD 0.00
ATPS4r (nd)
ADK1 1.00e+3
ATPM 8.39
O2t (nd)
EX_o2_e -14.4
CYTBD (nd)
EX_h_e 3.45
NADH16 (nd)
FRD7 (nd)
SUCDi 4.29
SUCOAS -4.10
SUCCt3 (nd)
EX_succ_e 0.00
SUCCt2_2 (nd)
FUM 4.62
MDH 4.64
MALS 0.0203
ME1 0.00
ME2 0.00
AKGDH 4.29
AKGt2r (nd)
EX_akg_e 0.00
GLUDy -3.48
GLUt2r (nd)
EX_glu__L_e 0.00
GLUSy 0.00
GLUN 0.00
GLNS 0.758
GLNabc (nd)
EX_gln__L_e 0.00
CS 4.69
ICL 0.00
CDHyr 4.69
ACONTa 4.69
ACONTb 4.69
Biomass_Ecoli_core_w_GAM (nd)

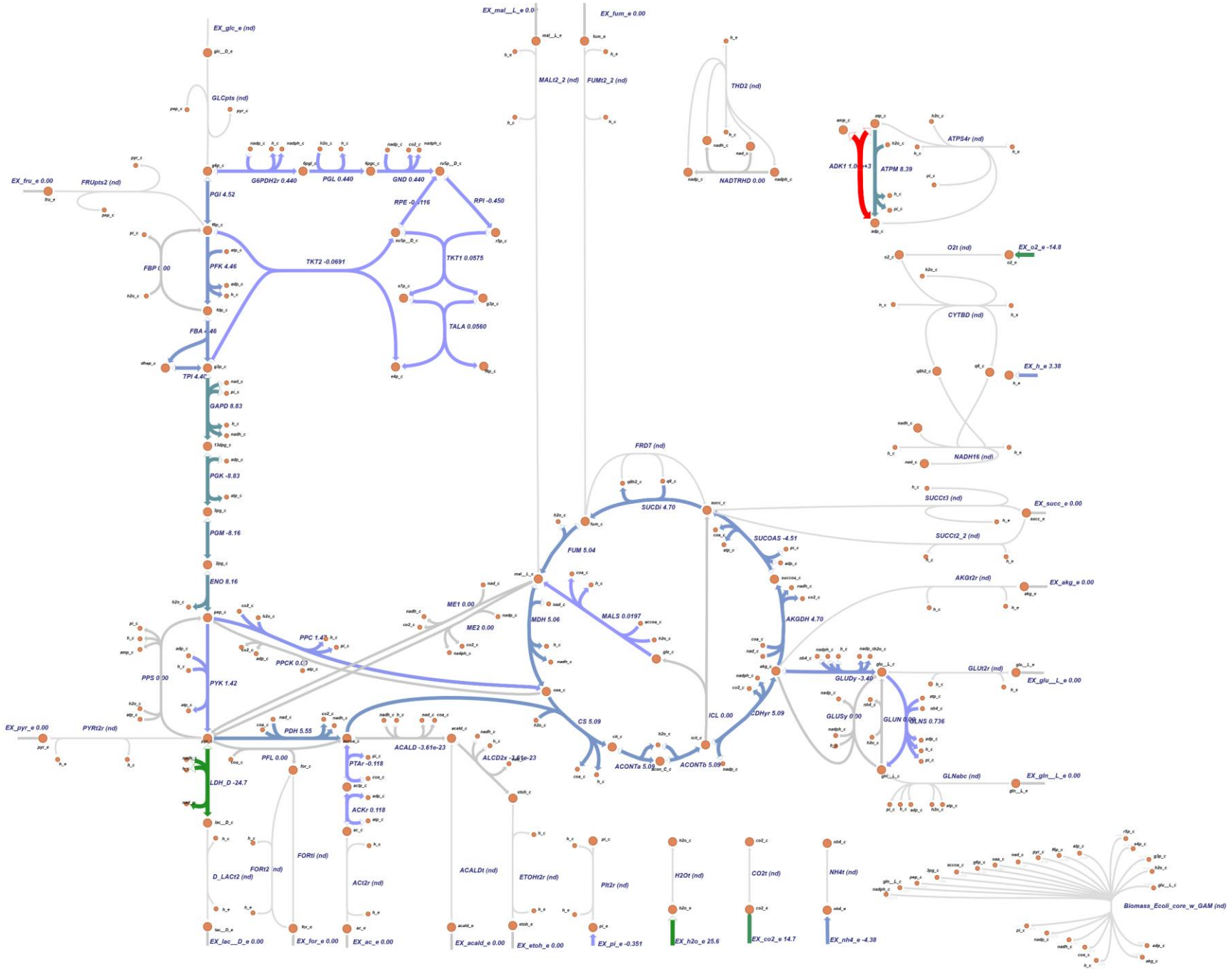
ΔlppA late


EX_glc_e (nd)
GLCpts (nd)
EX_fru_e 0.00
FRUpts2 (nd)
G6PDH2r 0.180
PGL 0.180
GND 0.180
PGI 4.78
RPE -0.192
RPI -0.370
TKT2 -0.161
TKT1 -0.0314
TALA -0.0329
PFK 4.53
FBP 0.00
FBA 4.53
TPI 4.48
GAPD 8.89
PGK -8.89
PGM -7.86
ENO 7.86
PPC 1.16
PPCK 0.00
PPS 0.00
PYK 1.42
EX_pyr_e 0.00
PYRt2r (nd)
PDH 0.00
PFL 5.53
LDH_D 0.00
D_LACt2 (nd)
FORt2 (nd)
FORti (nd)
ACt2r (nd)
PTAr -0.120
ACKr 0.120
ACALD 8.09e-25
ALCD2x 8.09e-25
ACALDt (nd)
ETOHt2r (nd)
EX_lac__D_e 0.00
EX_for_e 0.00
EX_ac_e 0.00
EX_acald_e 0.00
EX_etoh_e 0.00
EX_mal__L_e 0.00
EX_fum_e 0.00
MALt2_2 (nd)
FUMt2_2 (nd)
ME1 0.00
ME2 0.00
MDH 4.68
FUM 4.66
MALS 0.0202
CS 4.71
ACONTa 4.71
ACONTb 4.71
ICL 0.00
SUCDi 4.31
FRD7 (nd)
SUCOAS -4.12
AKGDH 4.31
GLDHyr 4.71
GLUDy -3.48
GLUSy 0.00
GLUN 0.00
GLNS 0.754
THD2 (nd)
NADTRHD 0.00
ATPS4r (nd)
ADK1 1.00e+3
ATPM 8.39
O2t (nd)
EX_o2_e -14.4
CYTBD (nd)
EX_h_e 3.46
NADH16 (nd)
SUCCt3 (nd)
EX_succ_e 0.00
SUCCt2_2 (nd)
AKGt2r (nd)
EX_akg_e 0.00
GLUt2r (nd)
EX_glu__L_e 0.00
GLNabc (nd)
EX_gln__L_e 0.00
H2Ot (nd)
EX_h2o_e 25.5
CO2t (nd)
EX_co2_e 14.3
NH4t (nd)
EX_nh4_e -4.49
Pit2r (nd)
EX_pi_e -0.359
Biomass_Ecoli_core_w_GAM (nd)

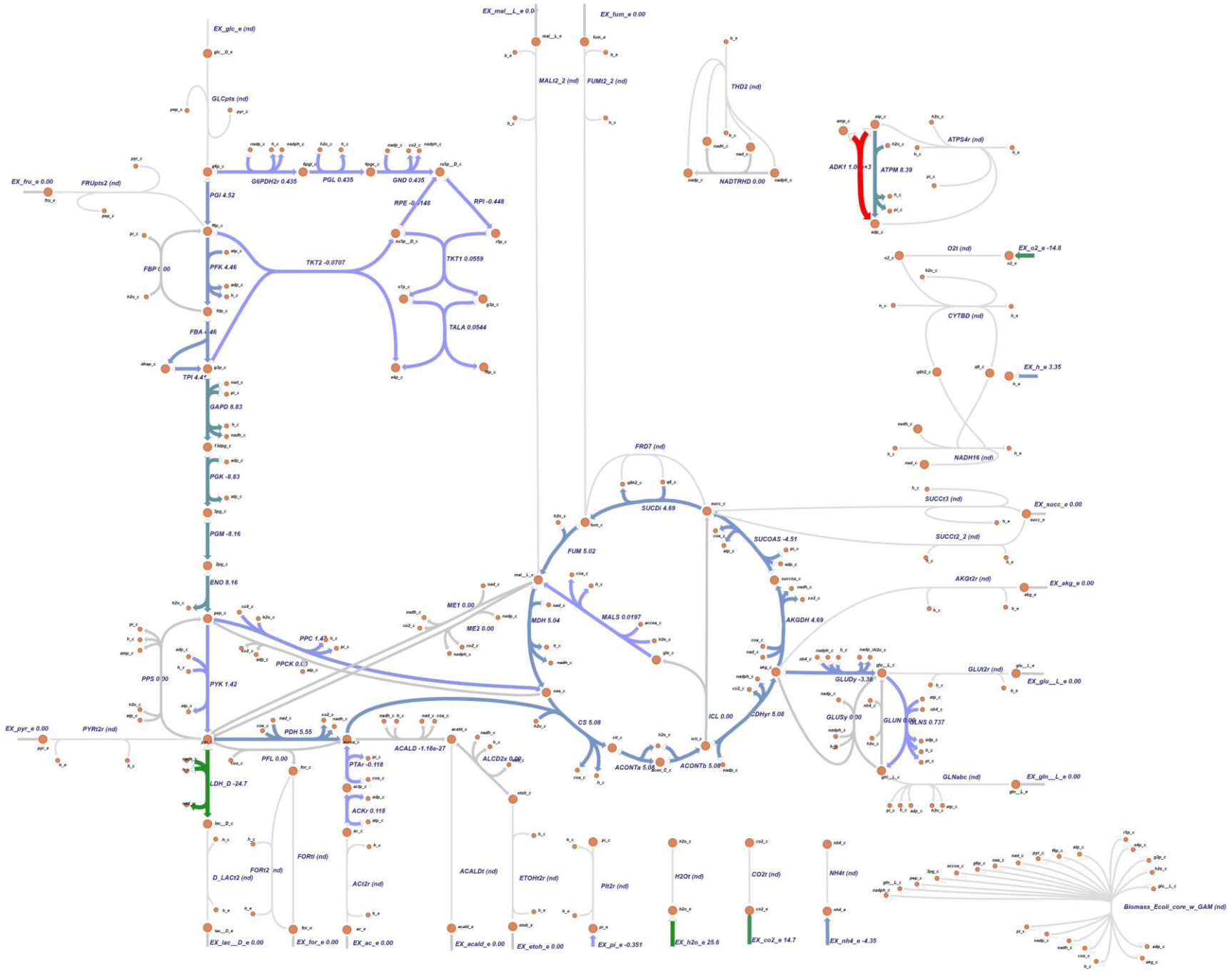
ΔlppB early


EX_glc_e (nd)
GLCpts (nd)
EX_fru_e 0.00
FRUpts2 (nd)
PGI 4.76
G6PDH2r 0.197
PGL 0.197
GND 0.197
RPE -0.082
RPI -0.377
TKT2 -0.156
TKT1 -0.0260
TALA -0.0276
PFK 4.53
FBA 4.53
TPI 4.47
GAPD 8.88
PGK -8.88
PGM -7.84
ENO 7.84
PPC 1.16
PPCK 0.00
PPS 0.00
PYK 1.40
EX_pyr_e 0.00
PYRt2r (nd)
PDH 0.00
PFL 5.51
LDH_D 0.00
D_LACt2 (nd)
FORt2 (nd)
FORti (nd)
ACt2r (nd)
PTAr -0.121
ACKr 0.121
ACALD -7.95e-21
ALCD2x 0.00
ACALDt (nd)
ETOHt2r (nd)
Pit2r (nd)
H2Ot (nd)
CO2t (nd)
NH4t (nd)
EX_lac__D_e 0.00
EX_for_e 0.00
EX_ac_e 0.00
EX_acald_e 0.00
EX_etoh_e 0.00
EX_pi_e -0.361
EX_h2o_e 25.5
EX_co2_e 14.3
EX_nh4_e -4.48
EX_mal__L_e 0.00
EX_fum_e 0.00
MALt2_2 (nd)
FUMt2_2 (nd)
THD2 (nd)
NADTRHD 0.00
ATPS4r (nd)
ADK1 1.00e+3
ATPM 8.39
O2t (nd)
EX_o2_e -14.4
CYTBD (nd)
EX_h_e 3.44
NADH16 (nd)
FRD7 (nd)
SUCDi 4.28
SUCCt3 (nd)
EX_succ_e 0.00
SUCCt2_2 (nd)
SUCOAS -4.09
FUM 4.62
AKGt2r (nd)
EX_akg_e 0.00
AKGDH 4.28
ME1 0.00
ME2 0.00
MDH 4.64
MALS 0.0203
CS 4.68
ACONTa 4.68
ACONTb 4.68
ICL 0.00
ICDHyr 4.68
GLUDy -3.48
GLUt2r (nd)
EX_glu__L_e 0.00
GLUSy 0.00
GLUN 0.00
GLNS 0.758
GLNabc (nd)
EX_gln__L_e 0.00
Biomass_Ecoli_core_w_GAM (nd)

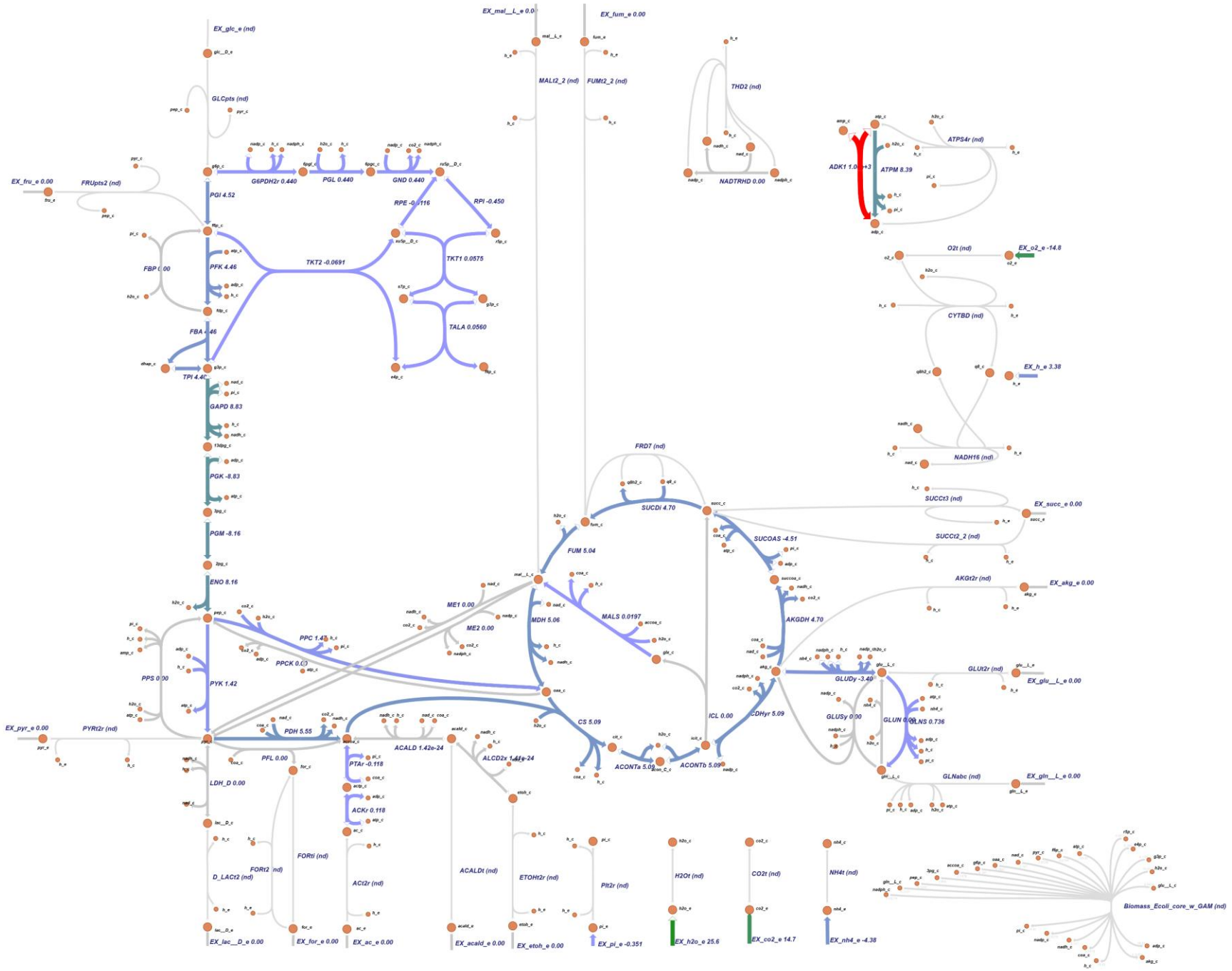
ΔlppB late


EX_glc_e (nd)
GLCpts (nd)
EX_fru_e 0.00
FRUpts2 (nd)
G6PDH2r 0.440
PGL 0.440
GND 0.440
PGI 4.52
RPE -0.116
RPI -0.450
TKT2 -0.0691
TKT1 0.0575
TALA 0.0560
PFK 4.46
FBA 4.46
TPI 4.46
GAPD 8.83
PGK -8.83
PGM -8.16
ENO 8.16
PPC 1.43
PPCK 0.00
PPS 0.00
PYK 1.42
EX_pyr_e 0.00
PYRtr2r (nd)
PDH 5.55
PFL 0.00
LDH_D -24.7
PTAr -0.118
ACKr 0.118
ACALD -3.61e-23
ALCD2x -3.61e-23
EX_mal__L_e 0.00
MALt2_2 (nd)
EX_fum_e 0.00
FUMt2_2 (nd)
THD2 (nd)
NADTRHD 0.00
ATPS4r (nd)
ADK1 1.00e+3
ATPM 8.39
O2t (nd)
EX_o2_e -14.8
CYTBD (nd)
EX_h_e 3.38
NADH16 (nd)
FRD7 (nd)
SUCDi 4.70
SUCOAS -4.51
SUCCt3 (nd)
EX_succ_e 0.00
SUCCt2_2 (nd)
FUM 5.04
MDH 5.06
MALS 0.0197
ME1 0.00
ME2 0.00
AKGt2r (nd)
EX_akg_e 0.00
AKGDH 4.70
GLUDy -3.40
GLUt2r (nd)
EX_glu__L_e 0.00
GLUSy 0.00
GLUN 0.00
GLNS 0.736
GLNabc (nd)
EX_gln__L_e 0.00
CS 5.09
ACONTa 5.09
ACONTb 5.09
ICL 0.00
ICDHyr 5.09
D_LACt2 (nd)
FORt2 (nd)
FORti (nd)
ACt2r (nd)
ACALDt (nd)
ETOHt2r (nd)
Pit2r (nd)
H2Ot (nd)
CO2t (nd)
NH4t (nd)
Biomass_Ecoli_core_w_GAM (nd)
EX_lac__D_e 0.00
EX_for_e 0.00
EX_ac_e 0.00
EX_acald_e 0.00
EX_etoh_e 0.00
EX_pi_e -0.351
EX_h2o_e 25.6
EX_co2_e 14.7
EX_nh4_e -4.38

WT early


EX_glc_e (nd)
GLCpts (nd)
EX_fru_e 0.00
FRUpts2 (nd)
G6PDH2r 0.435
PGL 0.435
GND 0.435
PGI 4.52
RPE -0.148
RPI -0.448
TKT2 -0.0707
TKT1 0.0559
TALA 0.0544
PFK 4.46
FBA 4.46
TPI 4.46
GAPD 8.83
PGK -8.83
PGM -8.16
ENO 8.16
PPC 1.43
PPCK 0.00
PPS 0.00
PYK 1.42
EX_pyr_e 0.00
PYRtr2r (nd)
PDH 5.55
PFL 0.00
LDH_D -24.7
EX_lac__D_e 0.00
D_LACt2 (nd)
FORt2 (nd)
FORti (nd)
EX_for_e 0.00
PTAr -0.118
ACKr 0.118
ACt2r (nd)
EX_ac_e 0.00
ACALD -1.18e-27
ALCD2x 0.00
ACALDt (nd)
EX_acald_e 0.00
ETOHt2r (nd)
EX_etoh_e 0.00
EX_mal__L_e 0.00
MALt2_2 (nd)
EX_fum_e 0.00
FUMt2_2 (nd)
ME1 0.00
ME2 0.00
MDH 5.04
MALS 0.0197
CS 5.08
ACONTa 5.08
ACONTb 5.08
ICL 0.00
FUM 5.02
FRD7 (nd)
SUCDi 4.69
SUCOAS -4.51
AKGDH 4.69
GLUDy -3.38
GLUSy 0.00
GLUN 0.00
GLNS 0.737
GLDHyr 5.08
THD2 (nd)
NADTRHD 0.00
ATPS4r (nd)
ADK1 1.0e+3
ATPM 8.39
O2t (nd)
EX_o2_e -14.8
CYTBD (nd)
EX_h_e 3.35
NADH16 (nd)
SUCCt3 (nd)
EX_succ_e 0.00
SUCCt2_2 (nd)
AKGt2r (nd)
EX_akg_e 0.00
GLUt2r (nd)
EX_glu__L_e 0.00
GLNabc (nd)
EX_gln__L_e 0.00
Pit2r (nd)
EX_pi_e -0.351
H2Ot (nd)
EX_h2o_e 25.6
CO2t (nd)
EX_co2_e 14.7
NH4t (nd)
EX_nh4_e -4.35
Biomass_Ecoli_core_w_GAM (nd)

WT late


EX_glc_e (nd)
GLCpts (nd)
EX_fru_e 0.00
FRUpts2 (nd)
G6PDH2r 0.440
PGL 0.440
GND 0.440
PGI 4.52
RPE -0.116
RPI -0.450
TKT2 -0.0691
TKT1 0.0575
TALA 0.0560
PFK 4.46
FBP 0.00
FBA 4.46
TPI 4.46
GAPD 8.83
PGK -8.83
PGM -8.16
ENO 8.16
PPC 1.43
PPCK 0.00
PPS 0.00
PYK 1.42
EX_pyr_e 0.00
PYRtr2r (nd)
PDH 5.55
PFL 0.00
LDH_D 0.00
EX_lac__D_e 0.00
D_LACt2 (nd)
FORt2 (nd)
FORti (nd)
EX_for_e 0.00
PTAr -0.118
ACKr 0.118
ACt2r (nd)
EX_ac_e 0.00
ACALD 1.42e-24
ALCD2x 1.42e-24
ACALDt (nd)
EX_acald_e 0.00
ETOHt2r (nd)
EX_etoh_e 0.00
EX_mal__L_e 0.00
MALt2_2 (nd)
EX_fum_e 0.00
FUMt2_2 (nd)
ME1 0.00
ME2 0.00
MDH 5.06
MALS 0.0197
FUM 5.04
CS 5.09
ACONTa 5.09
ACONTb 5.09
ICL 0.00
FRD7 (nd)
SUCDi 4.70
SUCOAS -4.51
AKGDH 4.70
GLDHyr 5.09
GLUDy -3.40
GLUSy 0.00
GLUN 0.00
GLNS 0.736
GLNabc (nd)
EX_gln__L_e 0.00
GLUt2r (nd)
EX_glu__L_e 0.00
AKGt2r (nd)
EX_akg_e 0.00
SUCCt3 (nd)
EX_succ_e 0.00
SUCCt2_2 (nd)
THD2 (nd)
NADTRHD 0.00
ATPS4r (nd)
ADK1 1.00e+3
ATPM 8.39
O2t (nd)
EX_o2_e -14.8
CYTBD (nd)
EX_h_e 3.38
NADH16 (nd)
Pit2r (nd)
EX_pi_e -0.351
H2Ot (nd)
EX_h2o_e 25.6
CO2t (nd)
EX_co2_e 14.7
NH4t (nd)
EX_nh4_e -4.38
Biomass_Ecoli_core_w_GAM (nd)